INCH-POUND

MIL-M-28787/526
29 March 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DIRECT MEMORY INTERFACE I MODULE, KEY CODE DWG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic direct memory interface I module.

1.2 Classification. This specification covers one class of module. Class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.877 failures per million operating hours for class I when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DWG.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A47	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B10	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	O	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D2	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D3-D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E16	Ioe	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA
F2	Iih	VII	X	-2.60	0.04	-2.60	0.04	-2.60	0.04	mA
F3-F12	Iih	VII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA
F13-F19	Iih	VII	X	-1.70	0.12	-1.70	0.12	-1.70	0.12	mA
F20	Iih	VII	X	-1.10	0.12	-1.10	0.12	-1.10	0.12	mA
F21	Iih	VII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA
G1	Iil	VIII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA
G2	Iil	VIII	X	-2.60	0.04	-2.60	0.04	-2.60	0.04	mA
G3-G12	Iil	VIII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA
G13-G19	Iil	VIII	X	-1.70	0.12	-1.70	0.12	-1.70	0.12	mA
G20	Iil	VIII	X	-1.10	0.12	-1.10	0.12	-1.10	0.12	mA
G21	Iil	VIII	X	-0.60	0.02	-0.60	0.02	-0.60	0.02	mA

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H1-H16	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J16	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L13	Vio	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M	PSFC	XII	S	2.4	4.5	2.4	4.5	2.4	4.5	uF
N1	Icc	XIII	X	77.5	186.0	77.5	186.0	77.5	186.0	mA
N2	Icc	XIII	X	-39.8	-19.5	-39.8	-19.5	-39.8	-19.5	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1	C1/o	XVII	Q	2	50	2	50	2	50	pF
R2	C1/o	XVII	Q	4	75	4	75	4	75	pF
R3-R9	C1/o	XVII	Q	2	50	2	50	2	50	pF
R10,R11	C1/o	XVII	Q	2	75	2	75	2	75	pF
R12,R13	C1/o	XVII	Q	4	100	4	100	4	100	pF
R14-R16	C1/o	XVII	Q	4	125	4	125	4	125	pF
R17	C1/o	XVII	Q	6	125	6	125	6	125	pF
R18,R19	C1/o	XVII	Q	4	125	4	125	4	125	pF
R20	C1/o	XVII	Q	4	100	4	100	4	100	pF
R21	C1/o	XVII	Q	2	75	2	75	2	75	pF
R22-R28	C1/o	XVII	Q	6	100	6	100	6	100	pF
R29	C1/o	XVII	Q	6	125	6	125	6	125	pF
R30	C1/o	XVII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
R31	C1/D	XVII	0	4	100	4	100	4	100	pF
R32	C1/D	XVII	0	6	100	6	100	6	100	pF
R33	C1/D	XVII	0	2	125	2	125	2	125	pF
R34-R36	C1/D	XVII	0	4	125	4	125	4	125	pF
R37	C1/D	XVII	0	2	125	2	125	2	125	pF
R38,R39	C1/D	XVII	0	4	125	4	125	4	125	pF
S1-S4	Vic latchup	XVIII	0	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTR	XIX	0	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/526-1 for class I. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc, and (V-) of -16.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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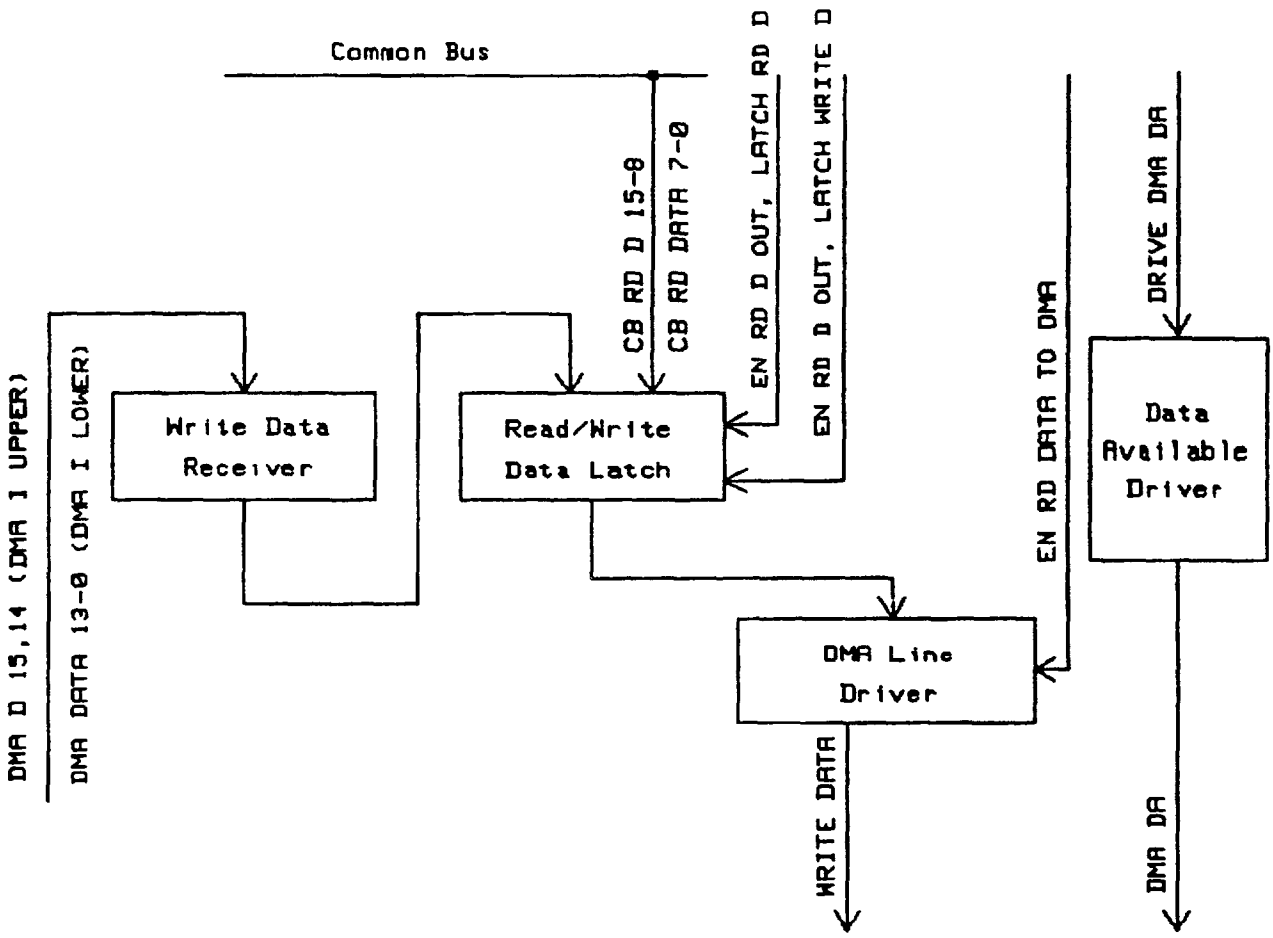


FIGURE 1. DWG functional block diagram.

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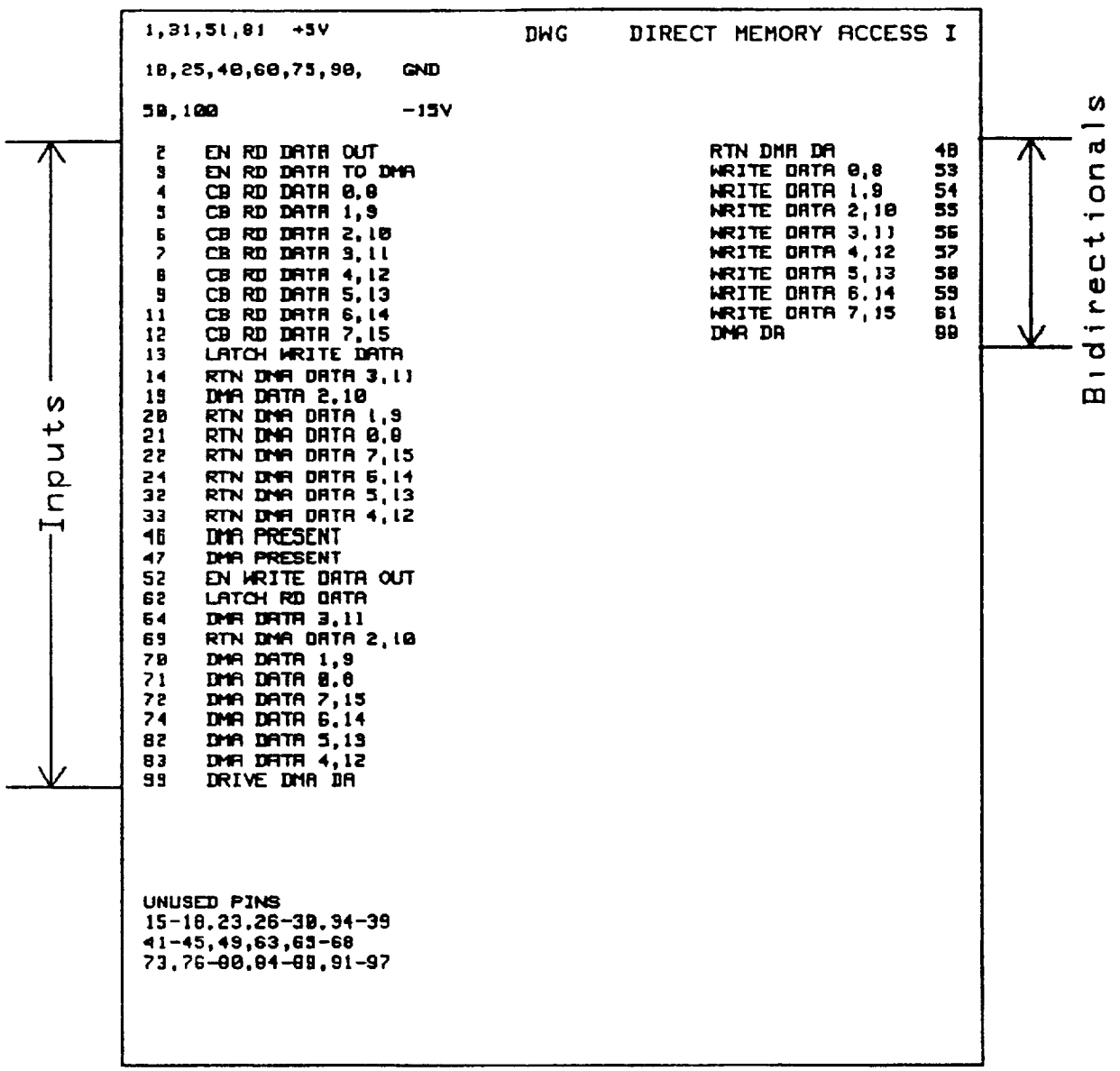


FIGURE 2. DWG pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L, or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

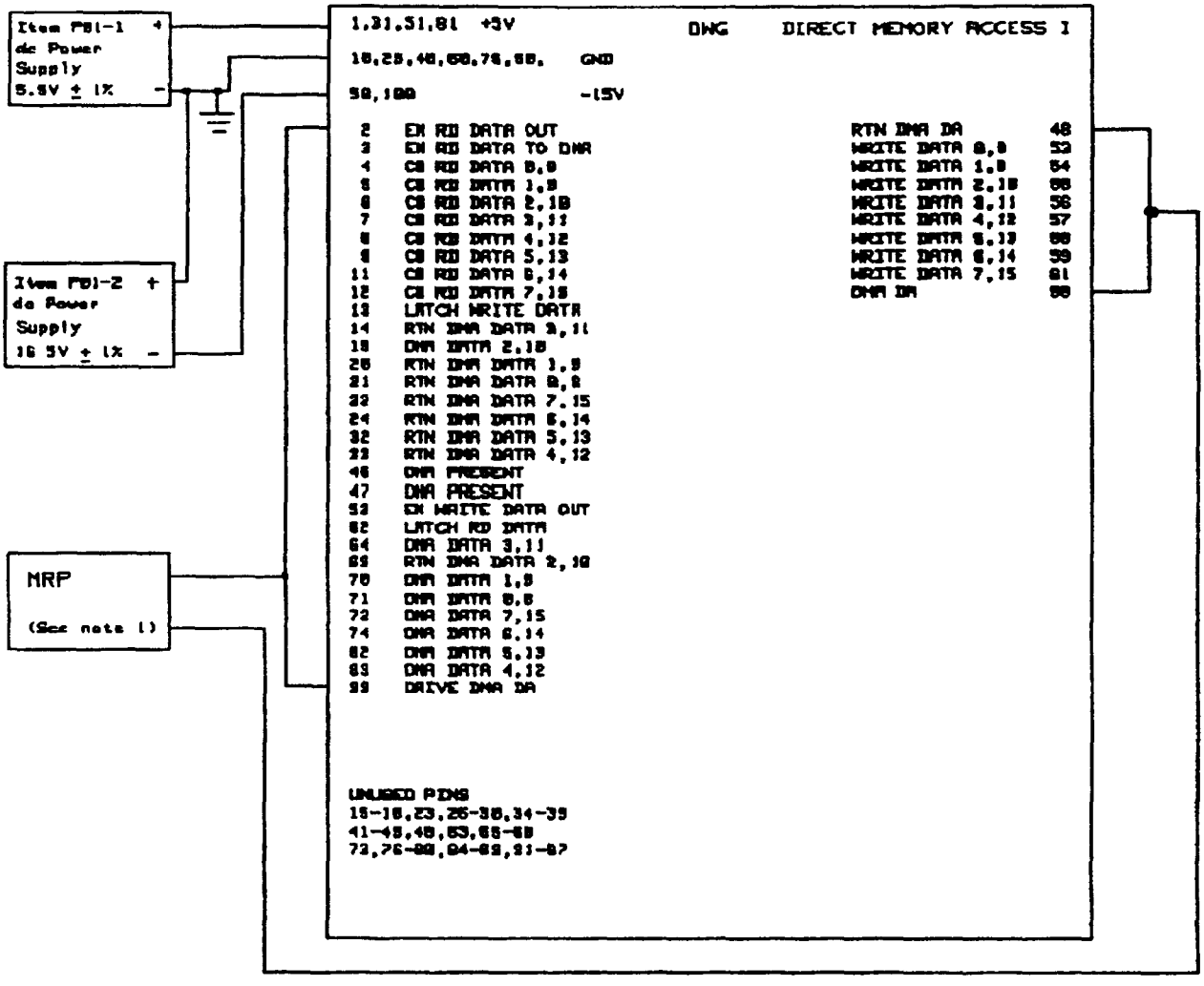
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. DWG operating temperature and component temperature test circuit.

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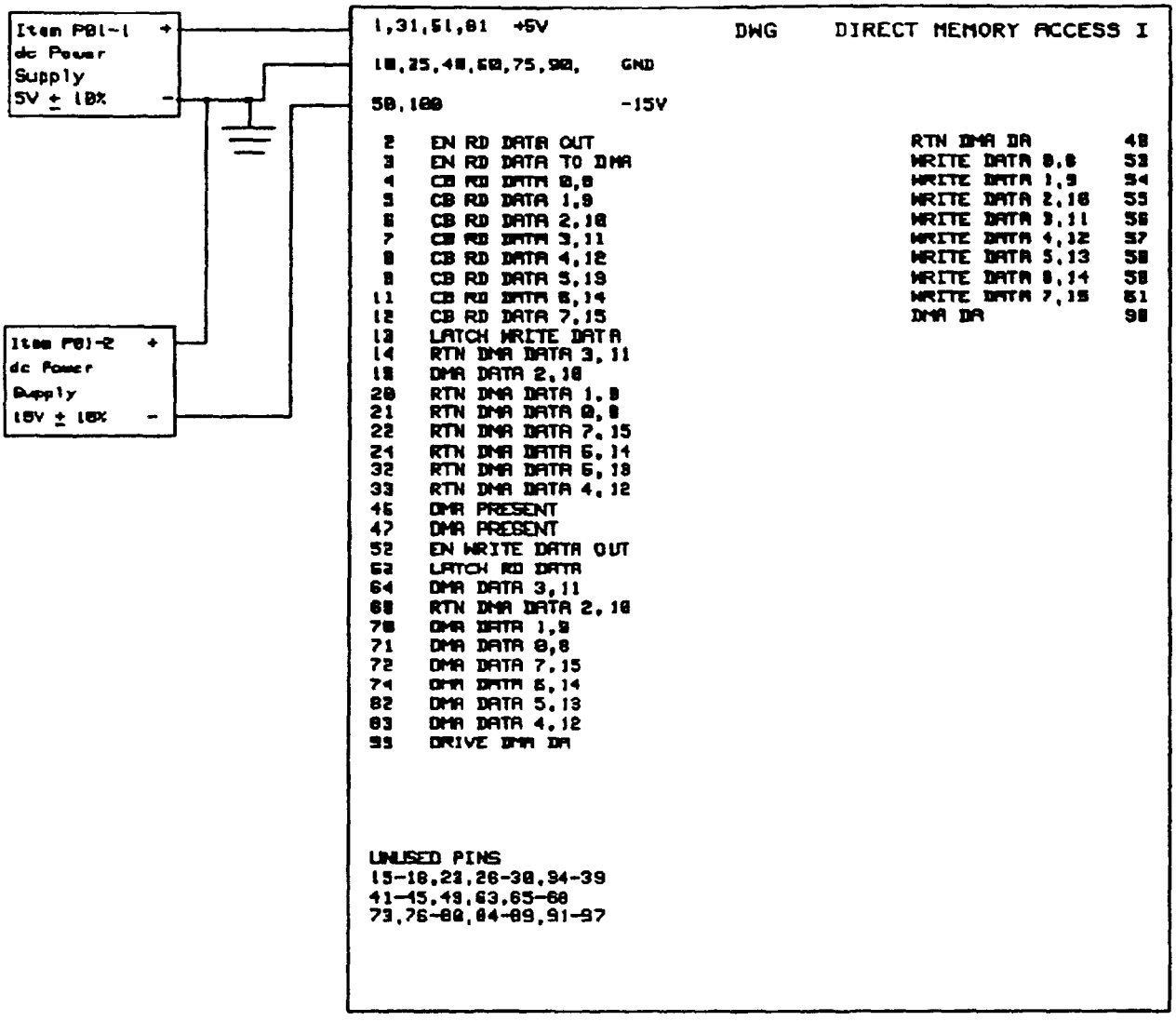


FIGURE 4. DWG vibration test circuit.

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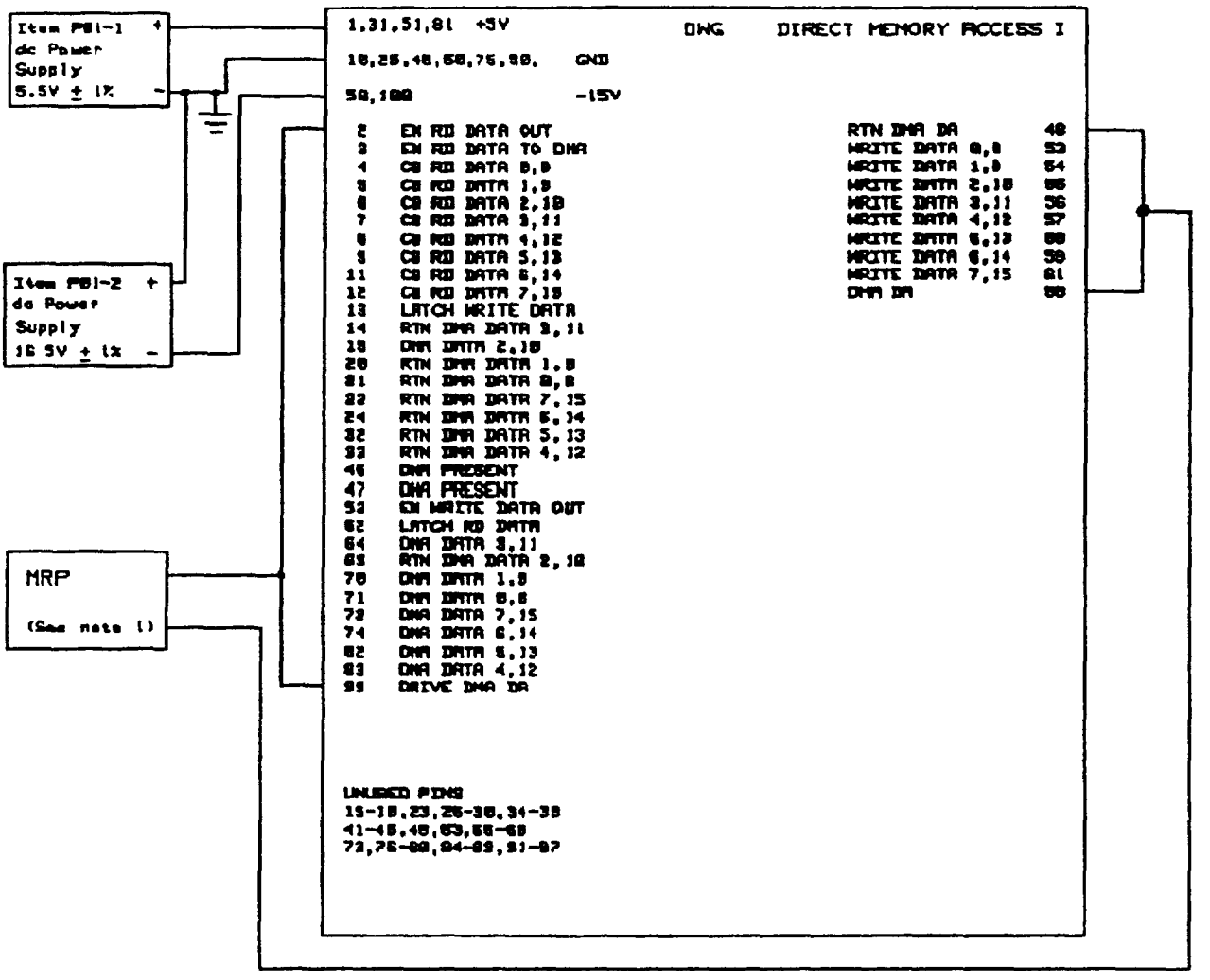
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. DWG life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 5V dc to 5.5 V dc, and (V-) -16.5V dc to -13.5V dc. Module power dissipation: 1.68 W (max); 1.15 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EN RD D OUT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
EN RD D DMA	-1.2	0.4	2.4	-2.6	0.04	0.2	75
CB RD D 0,8	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 1,9	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 2,10	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 3,11	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 4,12	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 5,13	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 6,14	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB RD D 7,15	-1.2	0.4	2.4	-0.6	0.02	0.1	75
L WRITE D	-1.2	0.4	2.4	-0.6	0.02	0.1	75
RTN DMA 3,11	NA	NA	NA	NA	NA	NA	100
DMA D 2,10	NA	NA	NA	NA	NA	NA	100
RTN DMA 1,9	NA	NA	NA	NA	NA	NA	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RTN DMA 0,8	NA	NA	NA	NA	NA	NA	125
RTN DMA 7,15	NA	NA	NA	NA	NA	NA	125
RTN DMA 6,14	NA	NA	NA	NA	NA	NA	125
RTN DMA 5,13	NA	NA	NA	NA	NA	NA	125
RTN DMA 4,12	NA	NA	NA	NA	NA	NA	125
DMA PRESENT	NA	NA	NA	NA	NA	NA	NA
DMA PRESENT	NA	NA	NA	NA	NA	NA	NA
EN WRITE OUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
LATCH RD D	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DMA D 3,11	NA	NA	NA	NA	NA	NA	100
RTN DMA 2,10	NA	NA	NA	NA	NA	NA	100
DMA D 1,9	NA	NA	NA	NA	NA	NA	125
DMA D 0,8	NA	NA	NA	NA	NA	NA	125
DMA D 7,15	NA	NA	NA	NA	NA	NA	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DMA D 6,14	NA	NA	NA	NA	NA	NA	125
DMA D 5,13	NA	NA	NA	NA	NA	NA	125
DMA D 4,12	NA	NA	NA	NA	NA	NA	125
DRIVE DMA DA	NA	NA	NA	NA	NA	NA	NA
RTN DMA DA	NA	NA	NA	NA	NA	NA	100
WRITE D 0,8	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 1,9	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 2,10	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 3,11	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 4,12	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 5,13	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 6,14	NA	0.4	2.4	-1.7	0.12	NA	100
WRITE D 7,15	NA	0.4	2.4	-1.7	0.12	NA	125
DMA DA	NA	NA	NA	NA	NA	NA	125

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
RTN DMA DA	NA	NA	NA	NA	NA	100
WRITE D 0,8	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 1,9	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 2,10	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 3,11	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 4,12	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 5,13	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 6,14	2.4	0.5	-0.4	6.0	-60.0	100
WRITE D 7,15	2.4	0.5	-0.4	6.0	-60.0	125
DMA DA	NA	NA	NA	NA	NA	125

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6.3.3 Class I dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
1	tpd	EN RD D DMA to DMA D 0,8	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 1,9	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 2,10	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 3,11	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 4,12	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 5,13	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 6,14	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 7,15	-	-	-	135	135	135	nS	11
2	tpd	DMA D 0,8 to WRITE D 0,8	-	-	-	88	88	88	nS	11
2	tpd	DMA D 1,9 to WRITE D 1,9	-	-	-	88	88	88	nS	11
2	tpd	DMA D 2,10 to WRITE D 2,10	-	-	-	88	88	88	nS	11
2	tpd	DMA D 3,11 to WRITE D 3,11	-	-	-	88	88	88	nS	11
2	tpd	DMA D 4,12 to WRITE D 4,12	-	-	-	88	88	88	nS	11

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6.3.3 Class I dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
2	tpd	DMA D 5,13 to WRITE D 5,13	-	-	-	88	88	88	nS	11
2	tpd	DMA D 6,14 to WRITE D 6,14	-	-	-	88	88	88	nS	11
2	tpd	DMA D 7,15 to WRITE D 7,15	-	-	-	88	88	88	nS	11
3	tpd	LATCH RD D to WRITE D 0,8	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 1,9	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 2,10	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 3,11	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 4,12	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 5,13	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 6,14	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 7,15	-	-	-	19	19	19	nS	11
4	tpd	DRIVE DMA DA to DMA DA	-	-	-	120	120	120	nS	11

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6.4 Module function. Direct Memory Access (DMA) is an optional feature that allows an external device access to main memory. It is composed of two data handling modules (DMA I, DWG) and two address/control modules (DMA II, DWH). The DMA I modules contain drivers and receivers for transfer of data to and from the DMA device via the DMA bus. Data that is written into main memory is sent from the receiver circuits on the DMA I modules to the common bus via the DMA II modules. Data read from main memory is transferred from the common bus via the DMA drivers to the external device. Control is exercised over the data transfers by the DMA II modules.

6.4.1 DMA operation. Three examples of DMA operation are described:

(a) DMA read operation
(b) DMA write operation
(c) DMA read/modify/write operation

6.4.1.1 DMA read operations. DMA read operations are initiated by the external device setting the read initiate line and activating the DMA control lines. When the DMA control logic on the DMA II module receives the read initiate, a request is made for common bus usage. The bus arbitrator module responds with the grant signal. The DMA II module then generates Output Ready (OR) and places the common bus function and address on the lines. Memory interface sends an Address Termination Ready (ATR), accepts the address and function bits and generates the Data Termination Ready (DTR). At this time, data is available from memory. The data is latched at the DMA I module and is available to the DMA device via the DMA bus when data available is active.

6.4.1.2 DMA write operations. DMA write operations are initiated by the external device setting the read initiate and DMA control lines applicable to the operation. When the DMA control logic receives read initiate, a common bus request is generated. The grant signal is received and OR, address, and function bits are placed on the lines. The ATR is received by the DMA II module and the address/function is accepted by the memory interface module. DMA data is placed on the common bus and is sampled when the second OR is generated by the DMA II module (after DTR generation). Data available acts as an acknowledge to the external device to clear the lines when the cycle is completed.

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6.4.1.3 DMA read/modify/write operations. DMA read/modify/write operations are initiated by the external device setting the read initiate line and DMA control lines. The DMA control logic receives the read initiate and a common bus request is initiated. Operation then proceeds per the read cycle previously discussed. Modifications are completed and the second part of the cycle is initiated by the DMA control lines. With the exception of the write initiate signal which initiates the write portion of the read/modify/write cycle, the balance of the write cycle is similar to the write cycle previously discussed. Data available acts as an acknowledge to the external device.

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6.4.2 Description of I/O pins.

Signal name	Pin number	I/O	Description
EN RD D OUT	2	I	EN RD DATA OUT
EN RD D DMA	3	I	EN RD DATA TO DMA
CB RD D 0,8	4	I	CB RD DATA 0,8
CB RD D 1,9	5	I	CB RD DATA 1,9
CB RD D 2,10	6	I	CB RD DATA 2,10
CB RD D 3,11	7	I	CB RD DATA 3,11
CB RD D 4,12	8	I	CB RD DATA 4,12
CB RD D 5,13	9	I	CB RD DATA 5,13
CB RD D 6,14	11	I	CB RD DATA 6,14
CB RD D 7,15	12	I	CB RD DATA 7,15
L WRITE D	13	I	LATCH WRITE DATA
RTN DMA 3,11	14	I	RTN DMA DATA 3,11
DMA D 2,10	19	I	DMA DATA 2,10
RTN DMA 1,9	20	I	RTN DMA DATA 1,9
RTN DMA 0,8	21	I	RTN DMA DATA 0,8
RTN DMA 7,15	22	I	RTN DMA DATA 7,15
RTN DMA 6,14	24	I	RTN DMA DATA 6,14
RTN DMA 5,13	32	I	RTN DMA DATA 5,13
RTN DMA 4,12	33	I	RTN DMA DATA 4,12
DMA PRESENT	46	I	DMA PRESENT

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DMA PRESENT	47	I	DMA PRESENT
EN WRITE OUT	52	I	EN WRITE DATA OUT
LATCH RD D	62	I	LATCH RD DATA
DMA D 3,11	64	I	DMA DATA 3,11
RTN DMA 2,10	69	I	RTN DMA DATA 2,10
DMA D 1,9	70	I	DMA DATA 1,9
DMA D 0,8	71	I	DMA DATA 0,8
DMA D 7,15	72	I	DMA DATA 7,15
DMA D 6,14	74	I	DMA DATA 6,14
DMA D 5,13	82	I	DMA DATA 5,13
DMA D 4,12	83	I	DMA DATA 4,12
DRIVE DMA DA	99	I	DRIVE DMA DA
RTN DMA DA	48	I/O	RTN DMA DA
WRITE D 0,8	53	I/O	WRITE DATA 0,8
WRITE D 1,9	54	I/O	WRITE DATA 1,9
WRITE D 2,10	55	I/O	WRITE DATA 2,10
WRITE D 3,11	56	I/O	WRITE DATA 3,11
WRITE D 4,12	57	I/O	WRITE DATA 4,12
WRITE D 5,13	58	I/O	WRITE DATA 5,13
WRITE D 6,14	59	I/O	WRITE DATA 6,14

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
WRITE D 7,15	61	I/O	WRITE DATA 7,15
DMA DA	98	I/O	DMA DA

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13,19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0072)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C, all unused pins	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	15,16,17, 18,23,26, 27,28,29, 30,34,35, 36,37,38, 39,41,42, 43,44,45, 49,63,65, 66,67,68, 73,76,77, 78,79,80, 84,85,86, 87,88,89, 91,92,93, 94,95,96, 97	10	INF	MOhm	1-8
A3	Isolation, unused to unused pins	15	16,17,18, 23,26,27, 28,29,30, 34,35,36, 37,38,39, 41,42,43, 44,45,49, 65,66,67, 68,73,76, 77,78,79, 80,84,85, 86,87,88, 89,91,92, 93,94,95, 96,97,63				
A4	Isolation, unused to unused pins	16	17,18,23, 26,27,28, 29,30,34,	10	INF	MOhm	1-8

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Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
			35,36,37, 38,39,41, 42,43,44, 45,49,65, 66,67,68, 73,76,77, 78,79,80, 81,85,86, 87,88,89, 91,92,93, 94,95,96, 97,63	10	INF	MOhm	1-8
A5	Isolation, unused to unused pins	17	18,23,26, 27,28,29, 30,34,35, 36,37,38, 39,41,42, 43,44,45, 49,63,65, 66,67,68, 73,76,77, 78,79,80, 84,85,86, 87,88,91, 92,93,94, 95,96,97				
A6	Isolation, unused to unused pins	18	23,26,27, 28,29,30, 34,35,36, 37,38,39, 41,42,43, 44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87,	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
			88,89,91, 92,93,94, 95,96,97	10	INF	MOhm	1-8
A7	Isolation, unused to unused pins	23	26,27,28, 29,30,34, 35,36,37, 38,39,41, 42,43,44, 45,49,65, 66,67,68, 73,76,77, 78,79,80, 84,85,86, 87,88,89, 91,92,93, 94,95,96, 97,63				
A8	Isolation, unused to unused pins	26	27,28,29, 30,34,35, 36,37,38, 39,41,42, 43,44,45, 49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63				
A9	Isolation, unused to unused pins	27	28,29,30, 34,35,36, 37,38,39, 41,42,43, 44,45,49,	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
			65,66,67, 68,73,76, 77,78,79, 80,84,85, 86,87,88, 89,91,92, 93,94,95, 96,97,63	10	INF	MOhm	1-8
A10	Isolation, unused to unused pins	28	29,30, 34,35,36, 37,38,39, 41,42,43, 44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97				
A11	Isolation, unused to unused pins	29	30,34,35, 36,37,38, 39,41,42, 43,44,45, 49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:34Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A12	Isolation, unused to unused pins	30	34,35,36, 37,38,39, 41,42,43, 44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,89, 91,92,93, 94,95,96, 97	10	INF	MOhm	1-8
A13	Isolation, unused to unused pins	34	35,36,37, 38,39,41, 42,43,44, 45,49,65, 66,67,68, 73,76,77, 78,79,80, 84,88,89, 91,92,93, 94,95,96, 97				
A14	Isolation, unused to unused pins	35	36,37,38, 39,41,42, 43,44,45, 49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:34Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A15	Isolation, unused to unused pins	36	37,38,39, 41,42,43, 44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,89, 91,92,93, 94,95,96, 97	10	INF	MOhm	1-8
A16	Isolation, unused to unused pins	37	38,39,41, 42,43,44, 45,49,65, 66,67,68, 73,76,77, 78,79,80, 84,88,89, 91,92,93, 94,95,96, 97				
A17	Isolation, unused to unused pins	38	39,41,42, 43,44,45, 49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A18	Isolation, unused to unused pins	39	41,42,43, 44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,89, 91,92,93, 94,95,96, 97	10	INF	MOhm	1-8
A19	Isolation, unused to unused pins	41	42,43,44, 45,49,65, 66,67,68, 73,76,77, 78,79,80, 84,88,89, 91,92,93, 94,95,96, 97				
A20	Isolation, unused to unused pins	42	43,44,45, 49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A21	Isolation, unused to unused pins	43	44,45,49, 63,65,66, 67,68,73, 76,77,78, 79,80,89, 91,92,93, 94,95,96, 97	10	INF	MOhm	1-8
A22	Isolation, unused to unused pins	44	45,49,65, 66,67,68, 73,76,77, 78,79,80, 84,88,89, 91,92,93, 94,95,96, 97				
A23	Isolation, unused to unused pins	45	49,65,66, 67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63				
A24	Isolation, unused to unused pins	49	65,66,67, 68,73,76, 77,78,79, 80,84,85, 86,87,88, 89,91,92, 93,94,95, 96,97,63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A25	Isolation, unused to unused pins	65	66,67,68, 73,76,77, 78,79,80, 84,85,86, 87,88, 89,91,92, 93,94,95, 96,97,63	10	INF	MOhm	1-8
A26	Isolation, unused to unused pins	66	67,68,73, 76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63				
A27	Isolation, unused to unused pins	67	68,73,76, 77,78,79, 80,84,85, 86,87,88, 89,91,92, 93,94,95, 96,97,63				
A28	Isolation, unused to unused pins	68	73,76,77, 78,79,80, 84,85,86, 87,88,89, 91,92,93, 94,95,96, 97,63				
A29	Isolation, unused to unused pins	73	76,77,78, 79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A30	Isolation, unused to unused pins	76	77,78,79, 80,84,85, 86,87,88, 91,92,93, 94,95,96, 97,63	10	INF	MOhm	1-8
A31	Isolation, unused to unused pins	77	78,79,80, 84,85,86, 87,88,89, 91,92,93, 94,95,96, 97,63				
A32	Isolation, unused to unused pins	78	79,80,84, 85,86,87, 88,89,91, 92,93,94, 95,96,97, 63				
A33	Isolation, unused to unused pins	79	80,84,85, 86,87,88, 89,91,92, 93,94,95, 96,97,63				
A34	Isolation, unused to unused pins	80	84,85,86, 87,88,89, 91,92,93, 94,95,96, 97,63				
A35	Isolation, unused to unused pins	84	85,86,87, 88,89,91, 92,93,94, 95,96,97, 63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A36	Isolation, unused to unused pins	85	86,87,88, 91,92,93, 94,95,96, 97,63	10	INF	MOhm	1-8
A37	Isolation, unused to unused pins	86	87,88,89, 91,92,93, 94,95,96, 97,63				
A38	Isolation, unused to unused pins	87	88,89, 91,92,93, 94,95,96, 97,63				
A39	Isolation, unused to unused pins	88	89,91,92, 93,94,95, 96,97,63				
A40	Isolation, unused to unused pins	89	91,92,93, 94,95,96, 97,63				
A41	Isolation, unused to unused pins	91	92,93,94, 95,96,97, 63				
A42	Isolation, unused to unused pins	92	93,94,95, 96,97,63				
A43	Isolation, unused to unused pins	93	94,95,96, 97,63				
A44	Isolation, unused to unused pins	95	96,97,63	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A45	Isolation, unused to unused pins	96	97,63	10	INF	MOhm	1-8
A46	Isolation, unused to unused pins	97	63	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A46
Isolation measurement	NA	± 500 KOhm	A1-A46

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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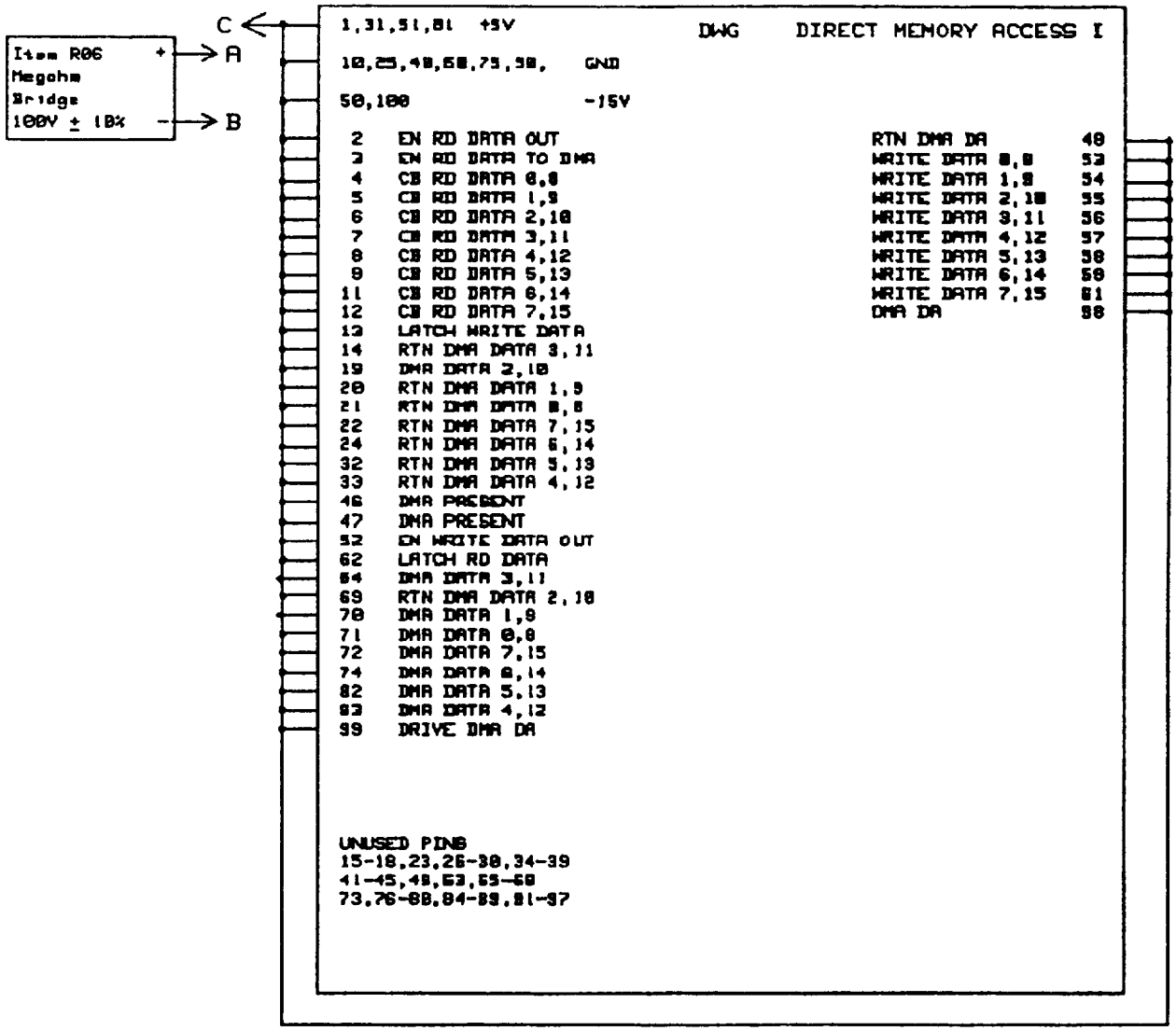


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	continuity, GND	10	60				
B4	Continuity, GND	60	75				
B5	Continuity, GND	60	90				
B6	Continuity, Vcc	1	31				
B7	Continuity, VCC	1	51				
B8	Continuity, Vcc	51	81				
B9	Continuity, -V	50	100				
B10	Continuity, DMA PRESENT	46	47	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B10

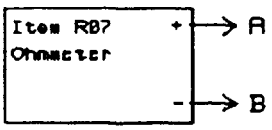
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,61,81 +5V

10,25,40,60,75,90, GND

50,100 -15V

DWG DIRECT MEMORY ACCESS I

2	EN RD DATA OUT	RTN DMA DA	48
3	EN RD DATA TO DMA	WRITE DATA 0,8	53
4	CB RD DATA 0,8	WRITE DATA 1,9	54
5	CB RD DATA 1,9	WRITE DATA 2,10	55
6	CB RD DATA 2,10	WRITE DATA 3,11	56
7	CB RD DATA 3,11	WRITE DATA 4,12	57
8	CB RD DATA 4,12	WRITE DATA 5,13	58
9	CB RD DATA 5,13	WRITE DATA 6,14	59
11	CB RD DATA 6,14	WRITE DATA 7,15	61
12	CB RD DATA 7,15	DMA DA	98
13	LATCH WRITE DATA		
14	RTN DMA DATA 3,11		
19	DMA DATA 2,10		
20	RTN DMA DATA 1,9		
21	RTN DMA DATA 0,8		
22	RTN DMA DATA 7,15		
24	RTN DMA DATA 6,14		
32	RTN DMA DATA 5,13		
33	RTN DMA DATA 4,12		
46	DMA PRESENT		
47	DMA PRESENT		
52	EN WRITE DATA OUT		
62	LATCH RD DATA		
64	DMA DATA 3,11		
69	RTN DMA DATA 2,10		
70	DMA DATA 1,9		
71	DMA DATA 0,8		
72	DMA DATA 7,15		
74	DMA DATA 6,14		
82	DMA DATA 5,13		
83	DMA DATA 4,12		
99	DRIVE DMA DA		

UNUSED PINS
15-18,23,26-30,34-38
41-45,49,63,65-68
73,76-80,84-89,91-97

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V-	16.5	10,25,40 60,75,90	50,100	Module shall pass all remaining tests	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	7.0 V -16.5 V	+0 mV,-70 mV +165mV,-0mv	C C
Power supply current limit (Vcc) (V-)	335 -72	NA NA	C C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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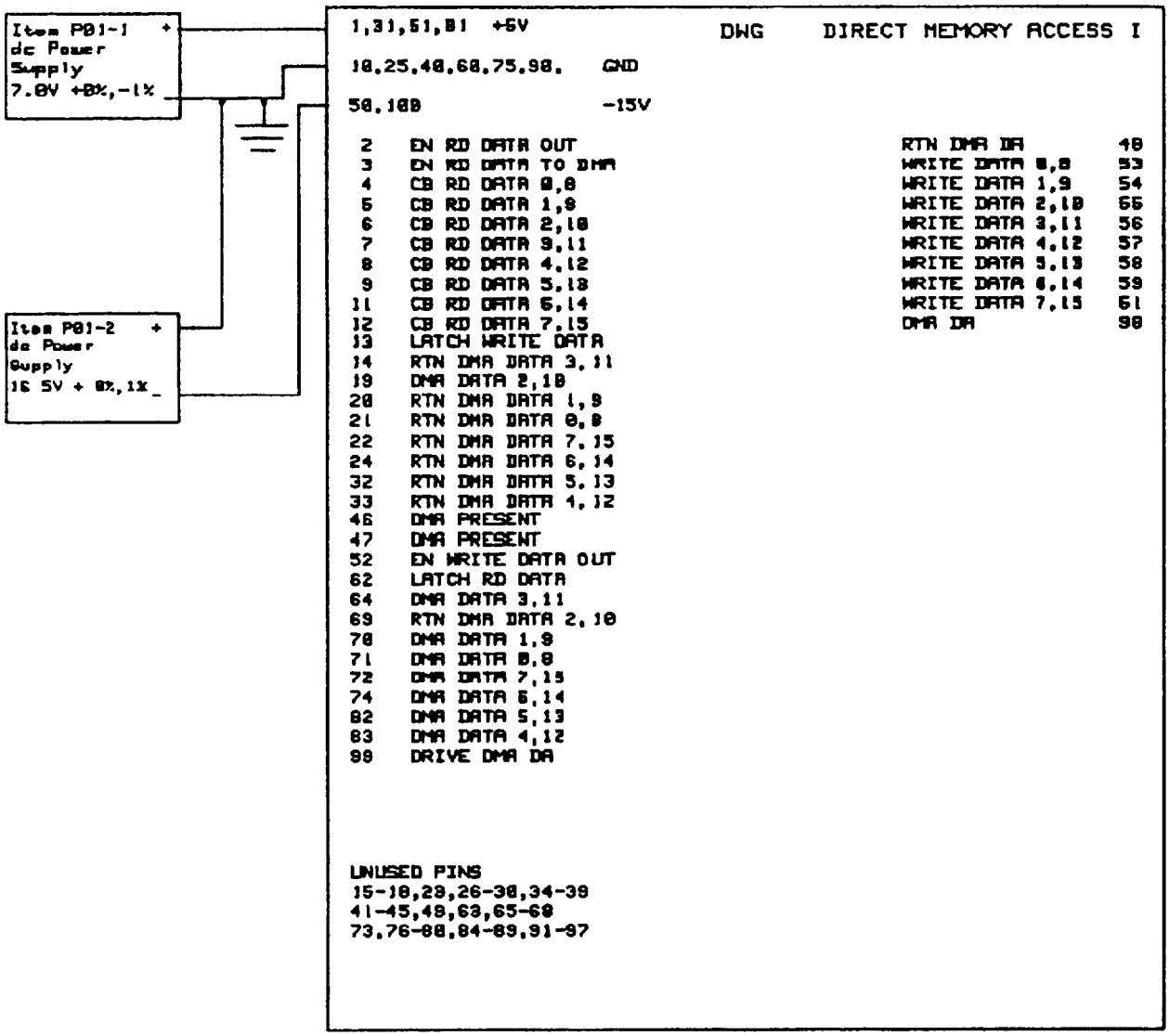


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	EN RD D OUT	2	5.5	0.0	0.1	mA	1-3
D2	EN RD D DMA	3	5.5	0.0	0.2	mA	1-3
D3	CB RD D 0,8	4	5.5	0.0	0.1	mA	1-3
D4	CB RD D 1,9	5					
D5	CB RD D 2,10	6					
D6	CB RD D 3,11	7					
D7	CB RD D 4,12	8					
D8	CB RD D 5,13	9					
D9	CB RD D 6,14	11					
D10	CB RD D 7,15	12					
D11	L WRITE D	13					
D12	EN WRITE OUT	52					
D13	LATCH RD D	62	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	5.5 V -16.5 V	± 55 mV ± 165 mV	D1-D13 D1-D13
Power supply current limit (Vcc) (V-)	279 mA -60 mA	NA NA	D1-D13 D1-D13
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D13
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D13 D1-D13
Input current measurement	NA NA NA	± 5 uA ± 10 uA ± 5 uA	D1 D2 D3-D13

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	WRITE D 0,8	53	0.0	-150.0	-60.0	mA	1-4
E2	WRITE D 0,8	53					
E3	WRITE D 1,9	54					
E4	WRITE D 1,9	54					
E5	WRITE D 2,10	55					
E6	WRITE D 2,10	55					
E7	WRITE D 3,11	56					
E8	WRITE D 3,11	56					
E9	WRITE D 4,12	57					
E10	WRITE D 4,12	57					
E11	WRITE D 5,13	58					
E12	WRITE D 5,13	58					
E13	WRITE D 6,14	59					
E14	WRITE D 6,14	59					
E15	WRITE D 7,15	61					
E16	WRITE D 7,15	61	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	5.5 V -16.5 V	± 55 mV ± 165 mV	E1-E16 E1-E16
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	E1-E16 E1-E16
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E16 E1-E16
Output current measurement	NA	± 4.5 mA	E1-E16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	EN RD D OUT	2	2.4	-0.60	0.02	mA	1-3
F2	EN RD D DMA	3	2.4	-2.60	0.04	mA	1-3
F3	CB RD D 0,8	4	2.4	-0.60	0.02	mA	1-3
F4	CB RD D 1,9	5					
F5	CB RD D 2,10	6					
F6	CB RD D 3,11	7					
F7	CB RD D 4,12	8					
F8	CB RD D 5,13	9					
F9	CB RD D 6,14	11					
F10	CB RD D 7,15	12					
F11	L WRITE D	13					
F12	EN WRITE OUT	52	2.4	-0.60	0.02	mA	1-3
F13	WRITE D 0,8	53	2.4	-1.70	0.12	mA	1-3
F14	WRITE D 1,9	54					
F15	WRITE D 2,10	55					
F16	WRITE D 3,11	56					
F17	WRITE D 4,12	57					
F18	WRITE D 5,13	58					
F19	WRITE D 6,14	59		-1.70			
F20	WRITE D 7,15	61	2.4	-1.10	0.12	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	LATCH RD D	62	2.4	-0.60	0.02	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F21
(V-)	-16.5 V	± 165 mV	F1-F21
Power supply current limit (Vcc)	279 mA	NA	F1-F21
(V-)	-60 mA	NA	F1-F21
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F21
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F21
low, (VL)	0.0 V	± 100 mV	F1-F21
Input current measurement	NA	± 31 uA	F1
	NA	± 132 uA	F2
	NA	± 31 uA	F3-F12
	NA	± 91 uA	F13-F20
	NA	± 31 uA	F21

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	EN RD D OUT	2	0.4	-0.60	0.02	mA	1-3
G2	EN RD D DMA	3	0.4	-2.60	0.04	mA	1-3
G3	CB RD D 0,8	4	0.4	-0.60	0.02	mA	1-3
G4	CB RD D 1,9	5					
G5	CB RD D 2,10	6					
G6	CB RD D 3,11	7					
G7	CB RD D 4,12	8					
G8	CB RD D 5,13	9					
G9	CB RD D 6,14	11					
G10	CB RD D 7,15	12					
G11	L WRITE D	13					
G12	EN WRITE OUT	52	0.4	-0.60	0.02	mA	1-3
G13	WRITE D 0,8	53	0.4	-1.70	0.12	mA	1-3
G14	WRITE D 1,9	54					
G15	WRITE D 2,10	55					
G16	WRITE D 3,11	56					
G17	WRITE D 4,12	57					
G18	WRITE D 5,13	58					
G19	WRITE D 6,14	59		-1.70			
G20	WRITE D 7,15	61	0.4	-1.10	0.12	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
G21	LATCH RD D	62	0.4	-0.60	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	5.5 V -16.5 V	± 55 mV ± 165 mV	G1-G21 G1-G21
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	G1-G21 G1-G21
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G21
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G21 G1-G21
Input current measurement	NA NA NA NA NA	± 31 uA ± 132 uA ± 31 uA ± 91 uA ± 31 uA	G1 G2 G3-G12 G13-G20 G21

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	WRITE D 0,8	53	-0.4	2.4	4.5	V	1-4
H2	WRITE D 0,8	53					
H3	WRITE D 1,9	54					
H4	WRITE D 1,9	54					
H5	WRITE D 2,10	55					
H6	WRITE D 2,10	55					
H7	WRITE D 3,11	56					
H8	WRITE D 3,11	56					
H9	WRITE D 4,12	57					
H10	WRITE D 4,12	57					
H11	WRITE D 5,13	58					
H12	WRITE D 5,13	58					
H13	WRITE D 6,14	59					
H14	WRITE D 6,14	59					
H15	WRITE D 7,15	61					
H16	WRITE D 7,15	61	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	4.5 V -13.5 V	± 45 mV ± 135 mV	H1-H16 H1-H16
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	H1-H16 H1-H16
Forcing current, (Ioh)	-0.4	± 10 % mA	H1-H16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H16 H1-H16
Output voltage measurement	NA	± 105 mV	H1-H16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

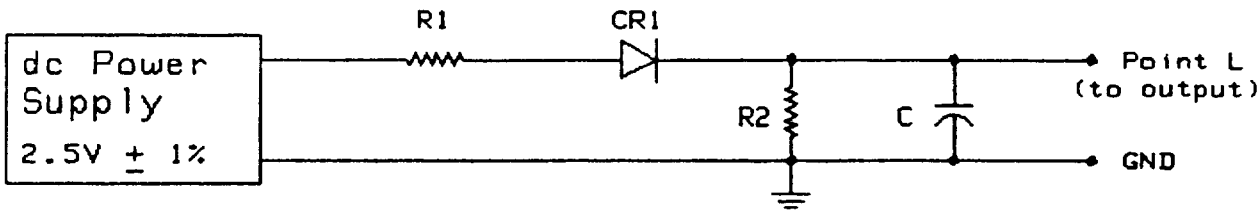
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
53,54,55,56,57, 58,59,61	6.2K	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	WRITE D 0,8	53	6.0	0.0	0.5	V	1-4
J2	WRITE D 0,8	53					
J3	WRITE D 1,9	54					
J4	WRITE D 1,9	54					
J5	WRITE D 2,10	55					
J6	WRITE D 2,10	55					
J7	WRITE D 3,11	56					
J8	WRITE D 3,11	56					
J9	WRITE D 4,12	57					
J10	WRITE D 4,12	57					
J11	WRITE D 5,13	58					
J12	WRITE D 5,13	58					
J13	WRITE D 6,14	59					
J14	WRITE D 6,14	59					
J15	WRITE D 7,15	61					
J16	WRITE D 7,15	61	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-)	4.5 V -13.5 V	± 45 mV ± 135 mV	J1-J16 J1-J16
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	J1-J16 J1-J16
Forcing current, (Iol)	6.0	600 uA	J1-J16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J16 J1-J16
Output voltage measurement	NA	± 25 mV	J1-J16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	EN RD D OUT	2	-18.0	-1.2	0.0	V	1-3
L2	EN RD D DMA	3					
L3	CB RD D 0,8	4					
L4	CB RD D 1,9	5					
L5	CB RD D 2,10	6					
L6	CB RD D 3,11	7					
L7	CB RD D 4,12	8					
L8	CB RD D 5,13	9					
L9	CB RD D 6,14	11					
L10	CB RD D 7,15	12					
L11	L WRITE D	13					
L12	EN WRITE OUT	52					
L13	LATCH RD D	62	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L13
Power supply current limit (Vcc) (V-)	NA NA	NA NA	L1-L13 L1-L13
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L13
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1-L13 L1-L13
Input voltage measurement	NA	± 60 mV	L1-L13

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	2.4	4.5	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 105 nF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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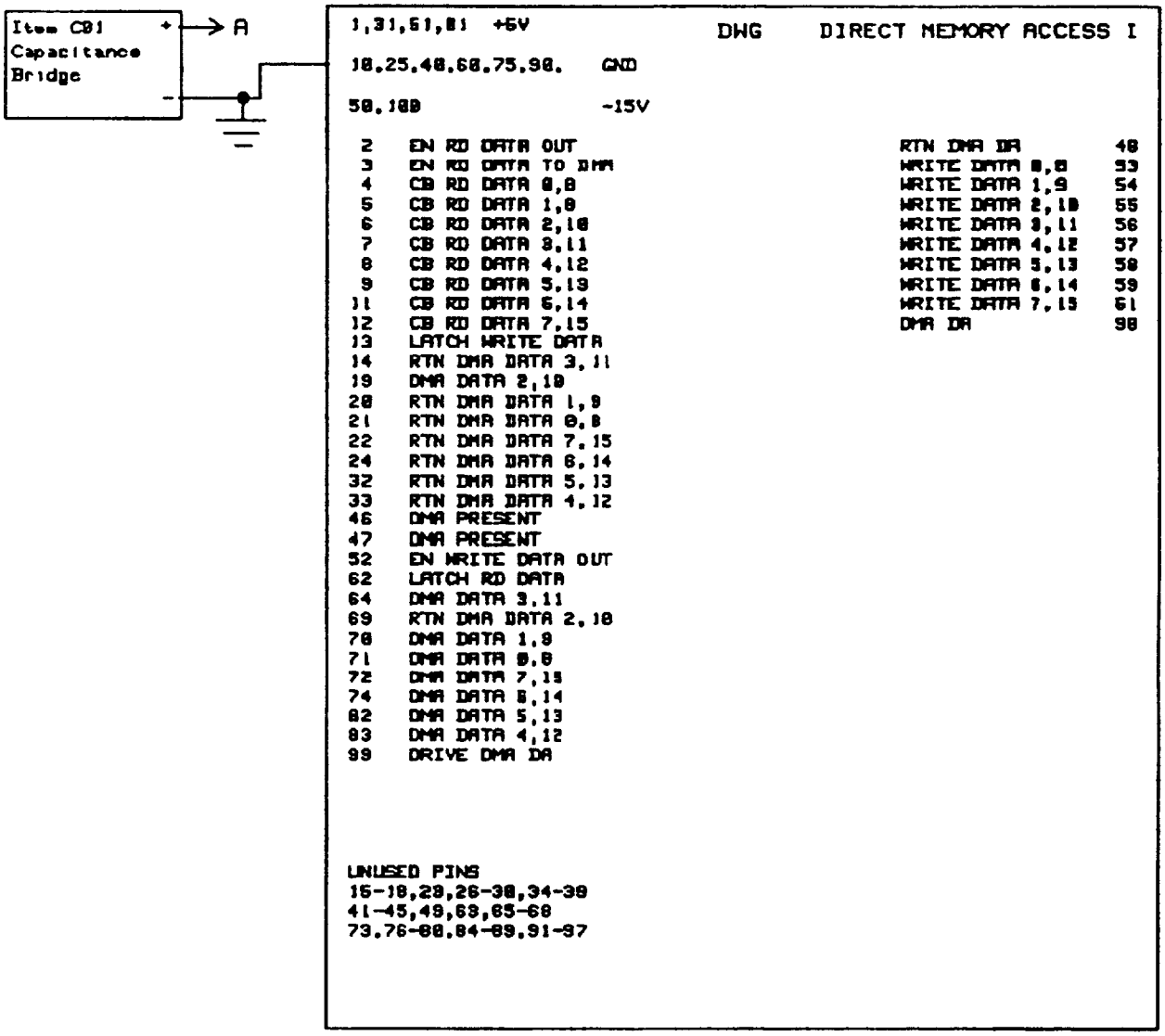


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc, +5V	1	77.5	186.0	mA	1-3
N2	Icc, -15V	50	-39.8	-19.5	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
(V-)	-16.5 V	± 165 mV	N1,N2
Power supply current limit, (Vcc)	279 mA	NA	N1,N2
(V-)	-60 mA	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 5.42 mA	N1
	NA	± 1.01 mA	N2

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2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (325 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (325 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V-) (Vcc)	4.5 V -15.0 V 5.5 V	± 45 mV ± 150 mV ± 55 mV	P1 P1,P2 P2
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	P1,P2 P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (325 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (325 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (325 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (325 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (Vcc) (V-)	4.5 V 5.5 V -15.0 V	± 45 mV ± 55 mV ± 150 mV	Q1,Q3 Q2,Q4 Q1-Q4
Power supply current limit, (Vcc) (V-)	279 mA -60 mA	NA NA	Q1-Q4 Q1-Q4
Input voltage high, (VH) low, (VL)	2.4 V 0.4 V	± 100 mV ± 100 mV	Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

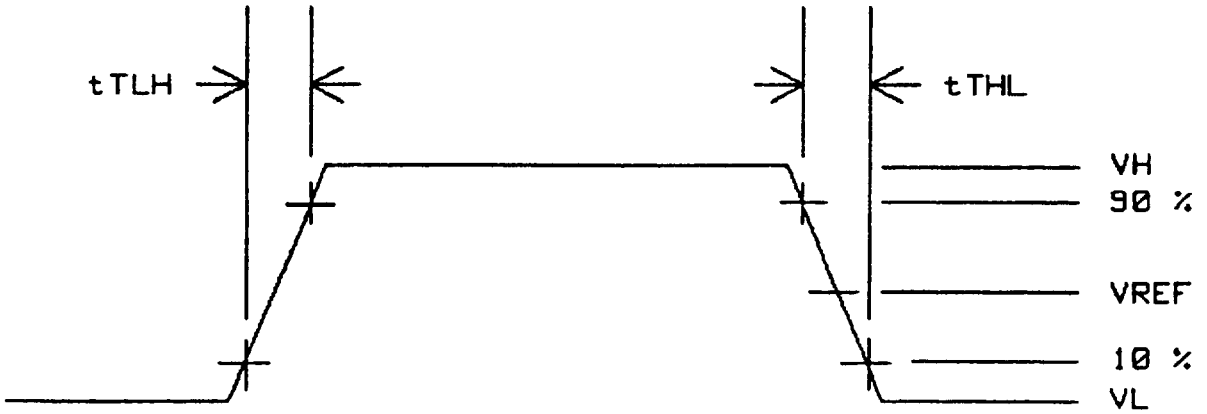
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
1	tpd	EN RD D DMA to DMA D 0,8	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 1,9	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 2,10	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 3,11	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 4,12	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 5,13	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 6,14	-	-	-	135	135	135	nS	11
1	tpd	EN RD D DMA to DMA D 7,15	-	-	-	135	135	135	nS	11
2	tpd	DMA D 0,8 to WRITE D 0,8	-	-	-	88	88	88	nS	11
2	tpd	DMA D 1,9 to WRITE D 1,9	-	-	-	88	88	88	nS	11
2	tpd	DMA D 2,10 to WRITE D 2,10	-	-	-	88	88	88	nS	11
2	tpd	DMA D 3,11 to WRITE D 3,11	-	-	-	88	88	88	nS	11
2	tpd	DMA D 4,12 to WRITE D 4,12	-	-	-	88	88	88	nS	11

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			0°C	25°C	60°C	0°C	25°C	60°C		
2	tpd	DMA D 5,13 to WRITE D 5,13	-	-	-	88	88	88	nS	11
2	tpd	DMA D 6,14 to WRITE D 6,14	-	-	-	88	88	88	nS	11
2	tpd	DMA D 7,15 to WRITE D 7,15	-	-	-	88	88	88	nS	11
3	tpd	LATCH RD D to WRITE D 0,8	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 1,9	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 2,10	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 3,11	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 4,12	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 5,13	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 6,14	-	-	-	19	19	19	nS	11
3	tpd	LATCH RD D to WRITE D 7,15	-	-	-	19	19	19	nS	11
4	tpd	DRIVE DMA DA to DMA DA	-	-	-	120	120	120	nS	11

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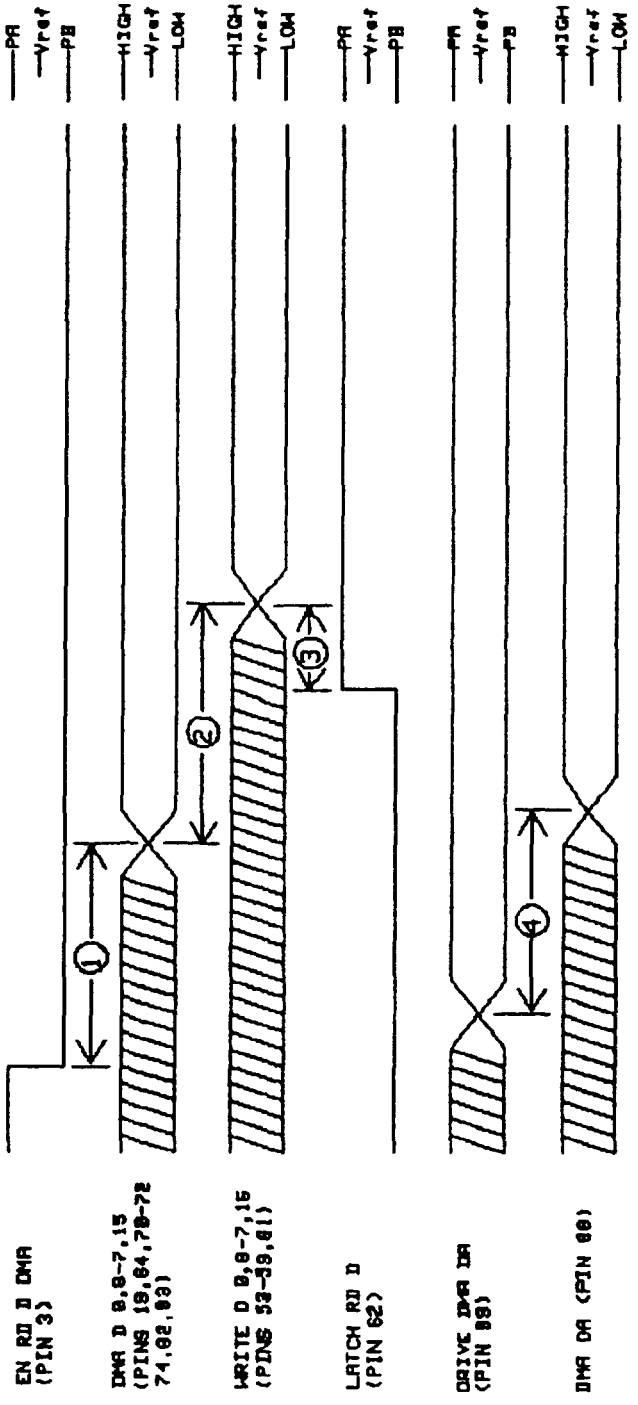


FIGURE 11. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	EN RD D OUT	2	2	50	pF	1-4
R2	EN RD D DMA	3	4	75	pF	1-4
R3	CB RD D 0,8	4	2	50	pF	1-4
R4	CB RD D 1,9	5				
R5	CB RD D 2,10	6				
R6	CB RD D 3,11	7				
R7	CB RD D 4,12	8				
R8	CB RD D 5,13	9				
R9	CB RD D 6,14	11	2	50	pF	1-4
R10	CB RD D 7,15	12	2	75	pF	1-4
R11	L WRITE D	13	2	75	pF	1-4
R12	RTN DMA 3,11	14	4	100	pF	1-4
R13	DMA D 2,10	19	4	100	pF	1-4
R14	RTN DMA 1,9	20	4	125	pF	1-4
R15	RTN DMA 0,8	21				
R16	RTN DMA 7,15	22	4	125	pF	1-4
R17	RTN DMA 6,14	24	6	125	pF	1-4
R18	RTN DMA 5,13	32	4	125	pF	1-4
R19	RTN DMA 4,12	33	4	125	pF	1-4
R20	RTN DMA DA	48	4	100	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	EN WRITE OUT	52	2	75	pF	1-4
R22	WRITE D 0,8	53	6	100	pF	1-4
R23	WRITE D 1,9	54				
R24	WRITE D 2,10	55				
R25	WRITE D 3,11	56				
R26	WRITE D 4,12	57				
R27	WRITE D 5,13	58				
R28	WRITE D 6,14	59	6	100	pF	1-4
R29	WRITE D 7,15	61	6	125	pF	1-4
R30	LATCH RD D	62	2	50	pF	1-4
R31	DMA D 3,11	64	4	100	pF	1-4
R32	RTN DMA 2,10	69	6	100	pF	1-4
R33	DMA D 1,9	70	2	125	pF	1-4
R34	DMA D 0,8	71	4	125	pF	1-4
R35	DMA D 7,15	72				
R36	DMA D 6,14	74	4	125	pF	1-4
R37	DMA D 5,13	82	2	125	pF	1-4
R38	DMA D 4,12	83	4	125	pF	1-4
R39	DMA DA	98	4	125	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R39
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R39
Capacitance measurement	NA	± 2.4 pF	R1
	NA	± 3.55 pF	R2
	NA	± 2.4 pF	R3-R9
	NA	± 3.65 pF	R10,R11
	NA	± 4.8 pF	R12,R13
	NA	± 6.05 pF	R14-R16
	NA	± 5.95 pF	R17
	NA	± 6.05 pF	R18,R19
	NA	± 4.8 pF	R20
	NA	± 3.65 pF	R21
	NA	± 4.7 pF	R22-R28
	NA	± 5.95 pF	R29
	NA	± 2.4 pF	R30
	NA	± 4.8 pF	R31
	NA	± 4.7 pF	R32
	NA	± 6.15 pF	R33
	NA	± 6.05 pF	R34-R36
	NA	± 6.15 pF	R37
	NA	± 6.05 pF	R38,R39

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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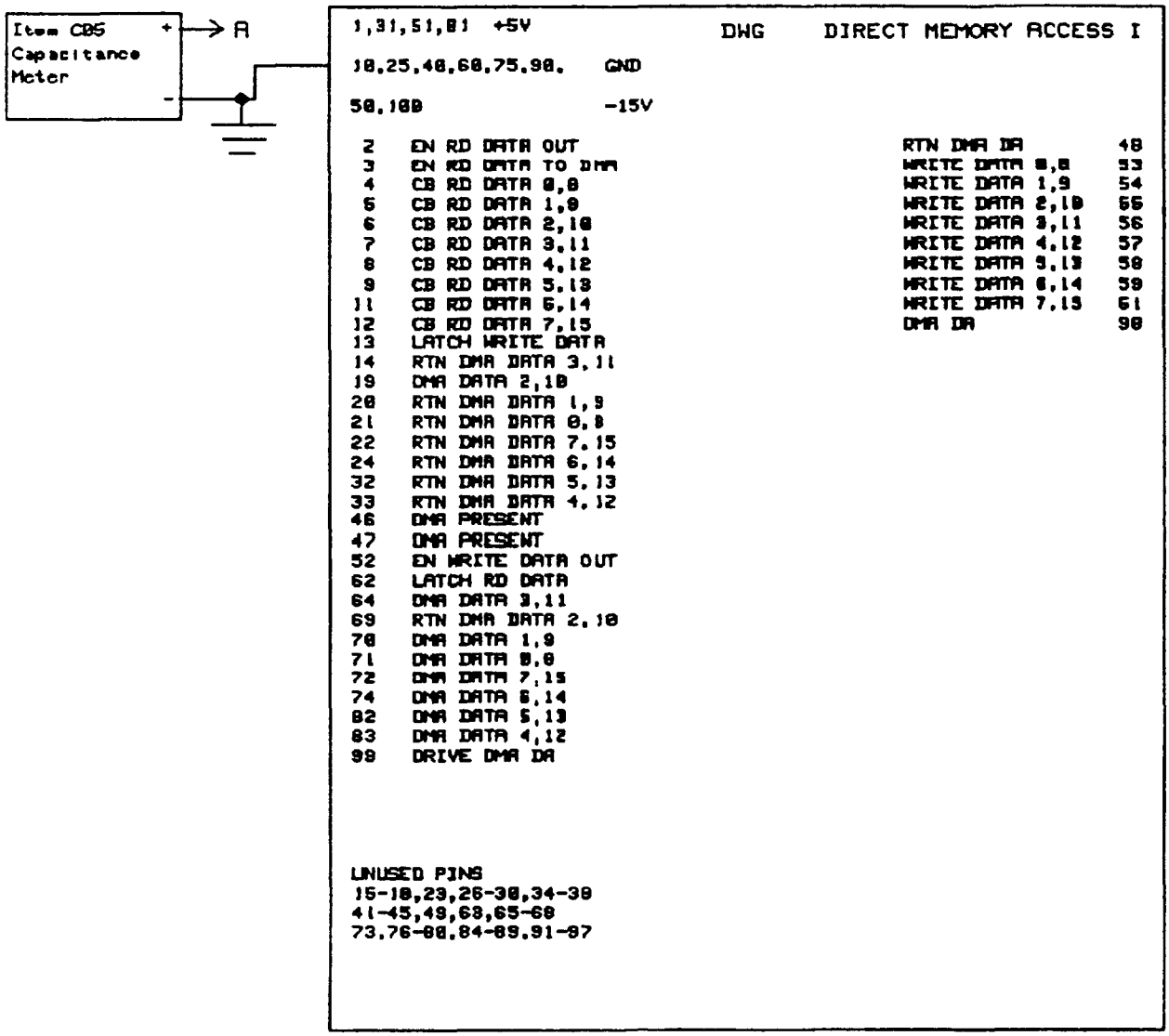


FIGURE 12. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Input clamp diode latchup test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S1	EN RD D OUT	2	-18.0	-1.2	2.4	V	1-3
S2	L WRITE D	13					
S3	EN WRITE OUT	52					
S4	LATCH RD D	62	-18.0	-1.2	2.4	V	1-3

NOTES:

1. Input clamp diode latchup test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S1-S4
(V-)	-13.5 V	±150 mV	S1-S4
Power supply current limit, (Vcc)	279 mA	NA	S1-S4
(V-)	-60 mA	NA	S1-S4
Forcing current, (Iin)	-18 mA	± 1.8 mA	S1-S4
Pattern voltage high, (VH)	NA	NA	S1-S4
low, (VL)	NA	NA	S1-S4
Input voltage measurement	NA	± 180 mV	S1-S4

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2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XVIII and the voltage then measured.

(d) Apply pulse to the module for a duration of 500 $\pm$ 20 nS.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA, Vcc	1	0	400	mV	1-4
T2	PSTA, V-	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
(V-)	-15.0 V	± 750 mV	T1,T2
Power supply current limit, (Vcc)	279 mA	NA	T1,T2
(V-)	-60 mA	NA	T1,T2
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1,T2
low, (VL)	0.0 V	± 100 mV	T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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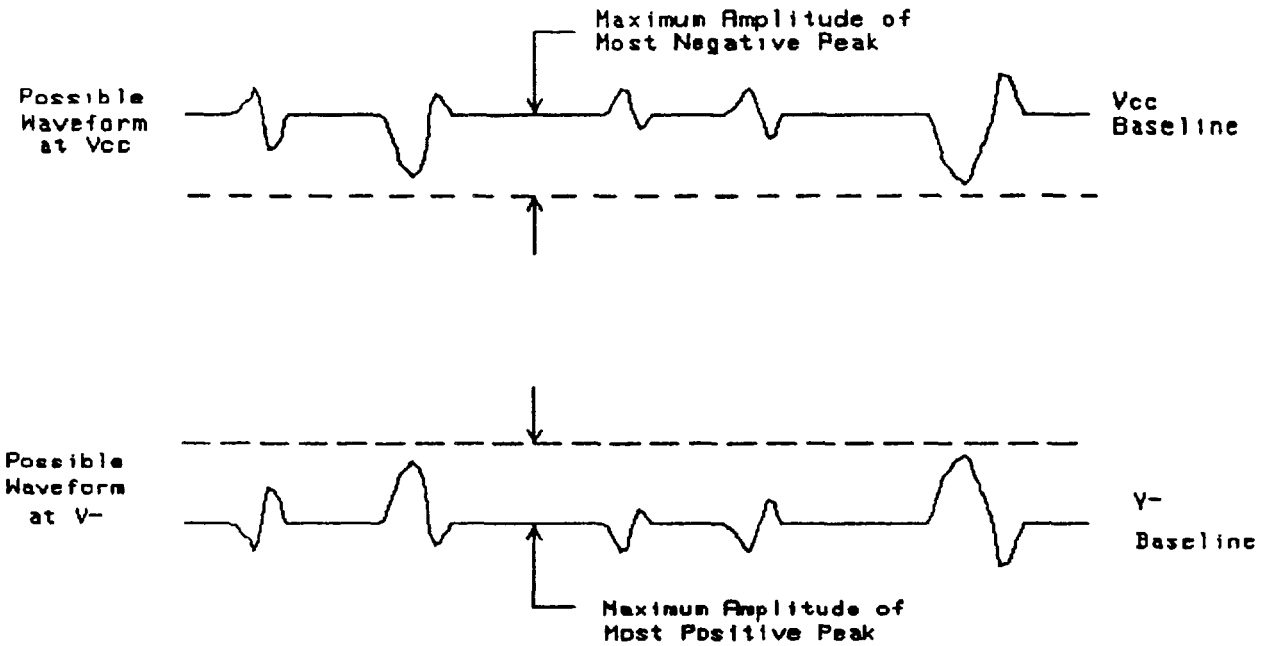
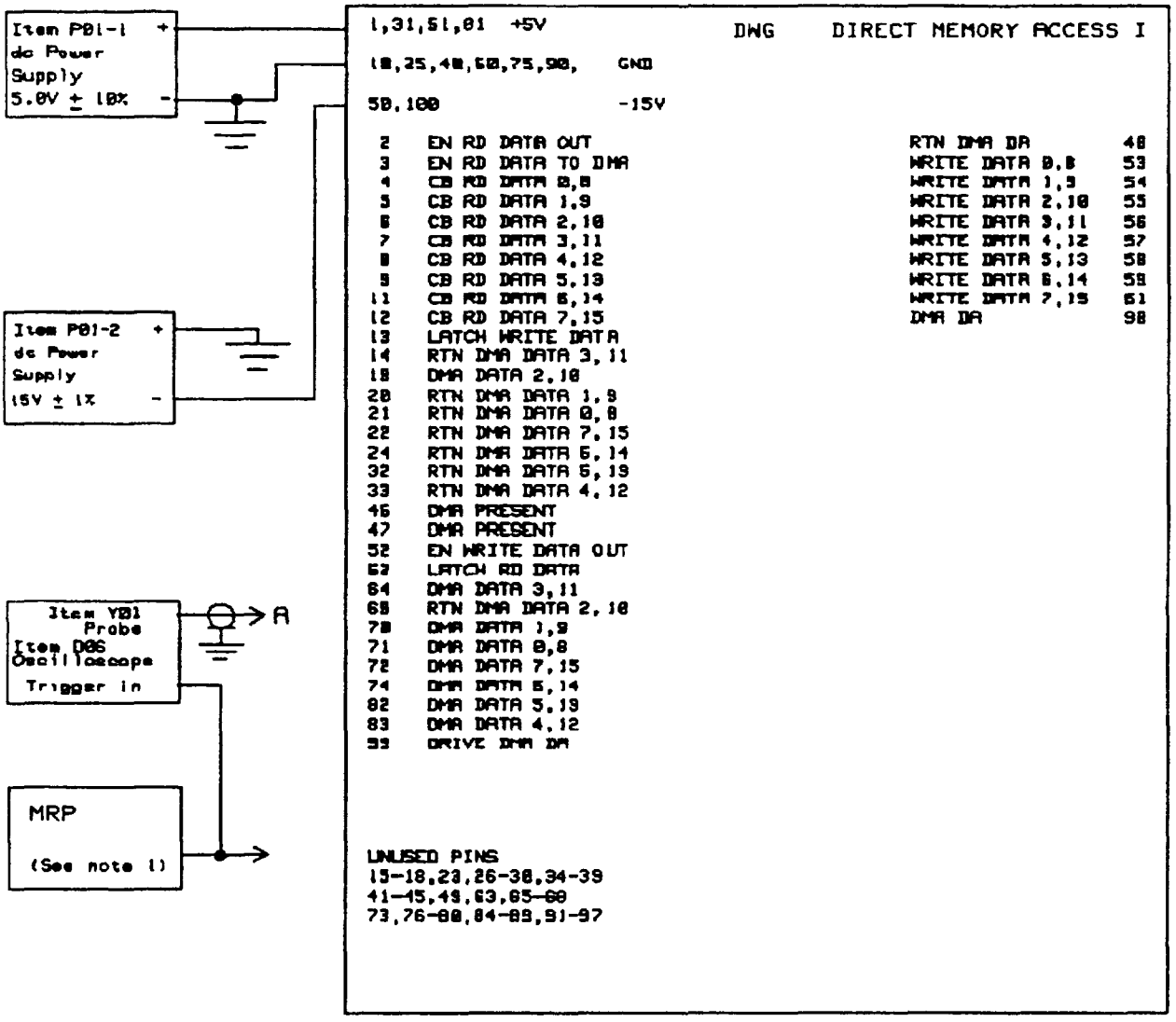


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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Modules, Standard Electronic, Direct Memory Interface I Module, Key Code DWG

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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